Executed test log

Alain Sauverain ISHEMA | ug2527343@ines.ac.rw | AI-A2-7343

Actual subprocess captures from tools/verify.py. Expected nonzero validation exits count as successful rejection checks. No skipped or failed runs are hidden from this log.

python run_all.py --student-id ug2527343@ines.ac.rw --seed 7343

Exit 0; expected 0

AI-A2-7343
SHA-256: 19a1558643adbce326107d7fcfb9df4a8cd48cf745f2af43b8e07af6b6975d62
regression LinearRegression {'mae': 0.7967032914337763, 'rmse': 0.9709049064598064, 'r2': 0.8702152124408429}
classification LogisticRegression {'precision': 0.9666666666666667, 'recall': 0.9354838709677419, 'f1': 0.9508196721311475, 'confusion_matrix': [[29, 1], [2, 29]], 'roc_auc': 0.989247311827957}
Selected K: 2 Silhouettes: {'2': 0.15658747030260753, '3': 0.13649622224245658, '4': 0.1360225628313596, '5': 0.13808832595085752, '6': 0.13715581796823964}
Rows: {'raw_rows': 247, 'clean_rows': 244, 'duplicate_rows': 3, 'missing_raw': {'rainfall_mm': 0, 'avg_temp_c': 0, 'soil_moisture_pct': 12, 'fertilizer_kg': 12, 'field_size_ha': 0, 'season': 0, 'cooperative_support': 0, 'yield_tons': 0, 'high_risk': 0}, 'missing_clean': {'rainfall_mm': 0, 'avg_temp_c': 0, 'soil_moisture_pct': 12, 'fertilizer_kg': 12, 'field_size_ha': 0, 'season': 0, 'cooperative_support': 0, 'yield_tons': 0, 'high_risk': 0}, 'train_rows': 183, 'test_rows': 61, 'split': '75/25 stratified high_risk; random_state=7343', 'row_id': 'zero-based original CSV data-row position'}
Generated artifacts: artifacts\classification_metrics.json, artifacts\clustering_report.json, artifacts\data_fingerprint.txt, artifacts\data_quality.json, artifacts\plots\clusters.png, artifacts\plots\confusion_matrix.png, artifacts\plots\residuals.png, artifacts\projection.json, artifacts\regression_metrics.json, artifacts\test_predictions.csv, models\classification.joblib, models\clustering.joblib, models\model_info.json, models\regression.joblib

[no stderr]

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[no stderr]

python predict.py --json samples/valid.json

Exit 0; expected 0

{
"predicted_yield_tons": 12.4816924638537,
"predicted_high_risk": 0,
"high_risk_probability": 0.33931241711599996,
"cluster": 0,
"verification_code": "AI-A2-7343"

}

[no stderr]

python predict.py --json samples/invalid.json

Exit 2; expected 2

[no stdout]

Input/model error: 3 validation errors for FarmRecord
rainfall_mm
Input should be less than or equal to 1400 [type=less_than_equal, input_value=1500, input_type=int]
For further information visit https://errors.pydantic.dev/2.14/v/less_than_equal
avg_temp_c
Input should be a valid number [type=float_type, input_value='warm', input_type=str]
For further information visit https://errors.pydantic.dev/2.14/v/float_type
season
Input should be 'A' or 'B' [type=literal_error, input_value='C', input_type=str]
For further information visit https://errors.pydantic.dev/2.14/v/literal_error

python run_all.py --student-id ug2527343@ines.ac.rw --seed 0

Exit 2; expected 2

[no stdout]

Error: Expected --student-id ug2527343@ines.ac.rw --seed 7343; incorrect identity or seed.

python -m pytest -q -ra

Exit 0; expected 0

..... [100%]
============================== warnings summary ===============================
.venv\lib\site-packages\fastapi\testclient.py:1
D:\AI\AI_A2_2527343\.venv\lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx` with `starlette.testclient` is deprecated; install `httpx2` instead.
from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
28 passed, 1 warning in 16.27s

[no stderr]

Reproducibility: exact byte equality for the fingerprint, metric JSONs, clustering report and prediction CSV across two full pipeline executions.

The first 22-test run also passed, with one Starlette/httpx deprecation warning. Its raw captures are retained in initial_execution_log.json. The suite was then expanded and rerun. Initial Matplotlib cache warnings were fixed by selecting a writable temporary cache. No failed or skipped pytest tests occurred.

Not performed: authentic personal recording, student signature, independent real-farm evaluation, production load testing and browser-driven manual Swagger interaction. API behavior was exercised with TestClient; this is not evidence of the student recording it.

Environment verification

Project-local virtual environment created empty before dependency installation; exact pins subsequently verified offline.

3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)]

Windows-10-10.0.26300-SP0

python -m pip install --no-index -r requirements.txt -> exit 0

Requirement already satisfied: annotated-doc==0.0.5 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 1)) (0.0.5)
Requirement already satisfied: annotated-types==0.8.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 2)) (0.8.0)
Requirement already satisfied: anyio==4.15.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 3)) (4.15.1)
Requirement already satisfied: certifi==2026.7.22 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 4)) (2026.7.22)
Requirement already satisfied: charset-normalizer==3.5.2 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 5)) (3.5.2)
Requirement already satisfied: click==8.5.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 6)) (8.5.0)
Requirement already satisfied: cloudpickle==3.1.2 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 7)) (3.1.2)
Requirement already satisfied: colorama==0.4.6 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 8)) (0.4.6)
Requirement already satisfied: contourpy==1.3.2 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 9)) (1.3.2)
Requirement already satisfied: cycler==0.12.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 10)) (0.12.1)
Requirement already satisfied: exceptiongroup==1.3.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 11)) (1.3.1)
Requirement already satisfied: fastapi==0.143.3 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 12)) (0.143.3)
Requirement already satisfied: fonttools==4.65.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 13)) (4.65.0)
Requirement already satisfied: h11==0.16.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 14)) (0.16.0)
Requirement already satisfied: httpcore==1.0.9 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 15)) (1.0.9)
Requirement already satisfied: httpx==0.28.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 16)) (0.28.1)
Requirement already satisfied: idna==3.20 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 17)) (3.20)
Requirement already satisfied: iniconfig==2.3.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 18)) (2.3.1)
Requirement already satisfied: joblib==1.6.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 19)) (1.6.0)
Requirement already satisfied: kiwisolver==1.5.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 20)) (1.5.1)
Requirement already satisfied: matplotlib==3.10.9 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 21)) (3.10.9)
Requirement already satisfied: numpy==2.2.6 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 22)) (2.2.6)
Requirement already satisfied: opentelemetry-api==1.45.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 23)) (1.45.1)
Requirement already satisfied: packaging==26.3 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 24)) (26.3)
Requirement already satisfied: pandas==2.3.3 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 25)) (2.3.3)
Requirement already satisfied: pillow==12.3.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 26)) (12.3.0)
Requirement already satisfied: pluggy==1.7.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 27)) (1.7.0)
Requirement already satisfied: pydantic==2.14.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 28)) (2.14.1)
Requirement already satisfied: pydantic_core==2.50.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 29)) (2.50.1)
Requirement already satisfied: Pygments==2.21.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 30)) (2.21.0)

Requirement already satisfied: pymupdf==1.28.2 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 31)) (1.28.2)
Requirement already satisfied: pyparsing==3.3.3 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 32)) (3.3.3)
Requirement already satisfied: pypdf==6.21.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 33)) (6.21.0)
Requirement already satisfied: pytest==9.1.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 34)) (9.1.1)
Requirement already satisfied: python-dateutil==2.9.0.post0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 35)) (2.9.0.post0)
Requirement already satisfied: pytz==2026.5 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 36)) (2026.5)
Requirement already satisfied: reportlab==5.0.1 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 37)) (5.0.1)
Requirement already satisfied: scikit-learn==1.7.2 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 38)) (1.7.2)
Requirement already satisfied: scipy==1.15.3 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 39)) (1.15.3)
Requirement already satisfied: six==1.17.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 40)) (1.17.0)
Requirement already satisfied: starlette==1.7.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 41)) (1.7.0)
Requirement already satisfied: threadpoolctl==3.7.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 42)) (3.7.0)
Requirement already satisfied: tomli==2.5.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 43)) (2.5.0)
Requirement already satisfied: typing-inspection==0.4.4 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 44)) (0.4.4)
Requirement already satisfied: typing_extensions==4.16.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 45)) (4.16.0)
Requirement already satisfied: tzdata==2026.5 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 46)) (2026.5)
Requirement already satisfied: uvicorn==0.54.0 in d:\ai\ai_a2_2527343\.venv\lib\site-packages (from -r requirements.txt (line 47)) (0.54.0)

[no stderr]

python -m pip check -> exit 0

No broken requirements found.

[no stderr]